ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

CCC 2123 : Operating Systems

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LAB PROJECT (20%)

KALI LINUX – FILE SYSTEM & OPERATIONS CHALLENGE

Lab Group : C1 (b)

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1.0 Introduction

This project demonstrates the creation and management of a University_Management system using the Kali Linux Command Line Interface (CLI). The directory structure simulates a university environment, including Courses, Attendance, Reports, and Tools, with multiple nested folders and files representing assignments, grades, syllabi, and automation scripts.

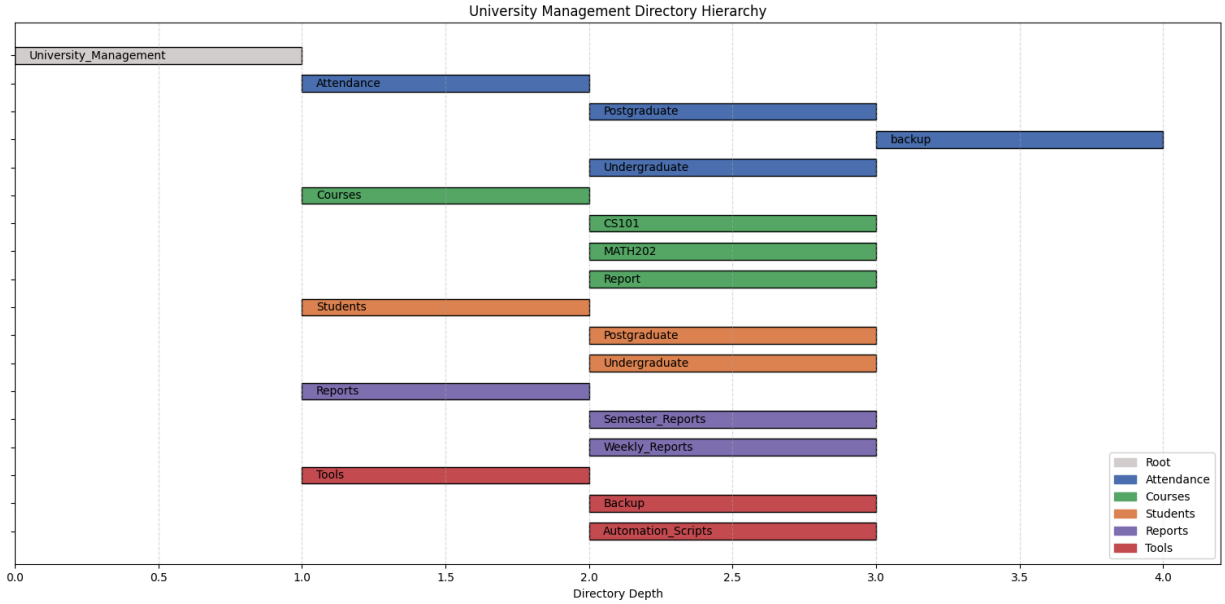
The project highlights the use of absolute and relative paths, file and directory manipulation, permission management, and automated task scheduling, showcasing practical CLI skills in organizing and maintaining a complex file system efficiently.

2.0 Objectives of the Assignment

The main objective of this project is to apply and reinforce practical **Kali Linux CLI skills** by designing and managing a **University_Management** system. The project aims to: organize a hierarchical directory structure, manipulate files and folders using both absolute and relative paths, control permissions, and automate recurring tasks with scripts. This helps demonstrate efficient file system management, content exploration, and automation techniques.

3.0 Directory Structure Visualization

The visualization provides a clear overview of the University Management directory hierarchy, showing how major components such as Students, Courses, Attendance, Reports, and Tools are logically structured. Color grouping is used to indicate related directories, making relationships between different sections easy to understand at a glance.



4.0 Description of Completed Tasks

The project was completed by successfully implementing all required tasks using the Kali Linux Command Line Interface within a structured **University Management** directory system.

Task 1: Directory & Project Structure Creation

A logical and hierarchical directory structure was designed to represent a university management system. Multiple top-level directories such as *Students, Courses, Attendance, Reports,* and *Tools* were created, along with several nested subdirectories. Files were generated at different directory levels using both absolute and relative paths, including creating files directly inside nested directories without changing into them.

Task 2: Directory Navigation Using Paths

Navigation through the project structure was performed using both absolute and relative paths. This included moving into nested directories, returning to parent directories, switching between the home directory and the project directory, and listing

directory contents without entering them. These actions demonstrated a clear understanding of Linux directory hierarchy and path usage.

Task 3: File Manipulation & Organization

Various file and directory operations were carried out to maintain an organized project structure. Files and folders were created, renamed, duplicated, moved, and removed from different directory levels. Reorganization was performed by relocating files into appropriate directories using correct path references, ensuring a clean and logical layout.

Task 4: Permissions & Ownership Management

File and directory permissions were modified using numeric and symbolic modes to control access for owners, groups, and others. Changes were applied using relative and absolute paths without navigating into each directory. Permission settings were verified using listing commands to confirm correct access control implementation.

Task 5: Exploration Tasks (Self-Directed Learning)

Two exploration tasks were completed:

- **File Exploration and Content Analysis:** Commands such as `find`, `grep`, `head`, and `less` were used to search for files, analyze content, and explore directory structures across multiple subdirectories.

- **Automation or Task Scheduling:** A shell script was created to automate report backups using archiving tools, and task scheduling was implemented using `cron` to execute the script automatically at defined intervals.

5.0 Task 1: Directory & Project Structure Creation

1- This command creates a new directory named University_Management inside the home directory using an absolute path, establishing the root folder for the project structure.

```
┌──(kali㉿kali)-[~]
└─$ mkdir /home/kali/University_Management
```

2- This command creates multiple top-level project directories (Students, Courses, Attendance, Reports, and Tools) inside the *University_Management* folder, forming the main components of the project structure using a relative path. The ls command then lists the contents, confirming the creation is done successfully.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ mkdir Students Cources Attendance Reports Tools
```

```
┌──(kali㉿kali)-[~/University_Management]
└─$ ls
Attendance  Courses  Reports  Students  Tools
```

3- These commands create two subdirectories (Undergraduate and Postgraduate) inside the Students directory using absolute paths. ls command for verification.

```
┌──(kali㉿kali)-[~/University_Management/Students]
└─$ mkdir /home/kali/University_Management/Students/Undergraduate

┌──(kali㉿kali)-[~/University_Management/Students]
└─$ mkdir /home/kali/University_Management/Students/Postgraduate

┌──(kali㉿kali)-[~/University_Management/Students]
└─$ ls
Postgraduate  Undergraduate
```

4- Created two subdirectories (CS101 and MATH202) inside the Courses directory using relative path.

```
┌──(kali㉿kali)-[~/University_Management/Courses]
└─$ mkdir CS101 MATH202
```

5- Created Assignments and Grades subdirectories inside each course folder (CS101 and MATH202) using relative path.

```
┌──(kali㉿kali)-[~/University_Management/Courses/CS101]
└─$ mkdir Assignments Grades
```

```
┌──(kali㉿kali)-[~/University_Management/Courses/MATH202]
└─$ mkdir Assignments Grades
```

6- This command creates Undergraduate and Postgraduate subdirectories inside the Attendance directory using relative path.

```
┌──(kali㉿kali)-[~/University_Management/Attendance]
└─$ mkdir Undergraduate Postgraduate
```

7- Created two subdirectories inside the `Reports` directory. The first command creates `Weekly_Reports`, and the second creates `Semester Reports`. Both use relative paths. ls to verify.

```
┌──(kali㉿kali)-[~]
└─$ mkdir University_Management/Reports/Weekly_Reports
```

```
┌──(kali㉿kali)-[~]
└─$ mkdir University_Management/Reports/Semester_Reports
```

```
┌──(kali㉿kali)-[~/University_Management/Reports]
└─$ ls
Semester_Reports  Weekly_Reports
```

8- Created a subdirectory inside the `Tools` directory, using relative path, then verify.

```
┌──(kali㉿kali)-[~/University_Management/Tools]
└─$ mkdir Automation_Scripts
```

```
┌──(kali㉿kali)-[~/University_Management/Tools]
└─$ ls
Automation_Scripts
```

9- Created individual student files using touch using absolute paths inside the Undergraduate and Postgraduate directories. Each file represents a student record. ls to verify.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ touch ~/University_Management/Students/Undergraduate/student_001.txt
touch ~/University_Management/Students/Undergraduate/student_002.txt
touch ~/University_Management/Students/Undergraduate/student_003.txt
touch ~/University_Management/Students/Postgraduate/student_101.txt
touch ~/University_Management/Students/Postgraduate/student_102.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Students/Undergraduate]
└─$ ls
student_001.txt  student_002.txt  student_003.txt
```

10- This set of commands creates course-related files for CS101, including a syllabus, assignment files, and grade records for students. The files are organized inside the respective Assignments and Grades subdirectories. These commands use absolute paths.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ touch ~/University_Management/Courses/CS101/Syllabus.txt
touch ~/University_Management/Courses/CS101/Assignments/Assignment1.txt
touch ~/University_Management/Courses/CS101/Assignments/Assignment2.txt
touch ~/University_Management/Courses/CS101/Grades/student_001.csv
touch ~/University_Management/Courses/CS101/Grades/student_002.csv
```

```
┌──(kali㉿kali)-[~]
└─$ ls /home/kali/University_Management/Courses/MATH202
Assignments  Grades  Syllabus.txt

┌──(kali㉿kali)-[~]
└─$ ls /home/kali/University_Management/Courses/CS101
Assignments  Grades  Syllabus.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Courses/CS101/Grades]
└─$ ls
student_001.csv  student_002.csv
```

11- Created a file named Assignments.txt inside the MATH202/Assignments directory and wrote the text into it. The command uses an absolute path. ls to verify the file exists.

```
┌──(kali㉿kali)-[~/University_Management/Courses/MATH202]
└─$  echo "This file is for the math subject assignments" > ~/University_Management/Courses
/MATH202/Assignments/Assignments.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Courses/MATH202/Assignments]
└─$ ls
Assignments.txt
```

12- Created file week1.txt inside both the Attendance/Undergraduate and Attendance/Postgraduate directories. Used absolute paths and verified.

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Undergraduate]
└─$ cat > ~/University_Management/Attendance/Undergraduate/week1.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Undergraduate]
└─$ ls
week1.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Postgraduate]
└─$ cat > ~/University_Management/Attendance/Postgraduate/week1.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Postgraduate]
└─$ ls
week1.txt
```

13- Created report_week1.txt inside Weekly_Reports, report_semester1.txt inside Semester_Reports, using absolute paths and verified it.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ touch ~/University_Management/Reports/Weekly_Reports/report_week1.txt
touch ~/University_Management/Reports/Semester_Reports/report_semester1.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Reports/Weekly_Reports]
└─$ ls
report_week1.txt
```

14- Created generate_reports.sh inside Tools/Automation_Scripts, using absolute paths and verified it.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ touch ~/University_Management/Tools/Automation_Scripts/generate_reports.sh
```

```
┌──(kali㉿kali)-[~/University_Management/Tools/Automation_Scripts]
└─$ ls
generate_reports.sh
```

15- Created a new directory 'emp_dir' inside Tools using an absolute path, and then creates an empty file 'unes_file.txt' inside the current directory (Tools) using a relative path

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p  /home/kali/University_Management/Tools/emp_dir
```

```
┌──(kali㉿kali)-[~/University_Management/Tools]
└─$ touch unes_file.txt
```

4.0 Task 2: Directory Navigation Using Paths

1- This sequence demonstrates directory navigation using relative paths to enter the `MATH202` course folder, move one level up to the parent directory, switch to the home directory, and then return to the project's root directory.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ cd Courses/MATH202

┌──(kali㉿kali)-[~/University_Management/Courses/MATH202]
└─$ cd ..

┌──(kali㉿kali)-[~/University_Management/Courses]
└─$ cd ~

┌──(kali㉿kali)-[~]
└─$ cd University_Management
```

2- Listed the contents of Courses and Attendance directories without entering them using a relative path then an absolute path.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ ls Courses

CS101  MATH202

┌──(kali㉿kali)-[~/University_Management]
└─$ ls /home/kali/University_Management/Attendance

Postgraduate  Undergraduate
```

3- Listed the contents of a deep directory MATH202 directly, using a relative path.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ ls Courses/MATH202

Assignments  Grades  Syllabus.txt
```

4- Moved back two levels in the hierarchy after entering MATH202 using a relative path.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ cd Courses/MATH202


┌──(kali㉿kali)-[~/University_Management/Courses/MATH202]
└─$ cd ../..

┌──(kali㉿kali)-[~/University_Management]
└─$
```

5- Entering a deep directory Weekly_Reports from home using an absolute path then return to the home again.

```
┌──(kali㉿kali)-[~]
└─$ /home/kali/University_Management/Reports/Weekly_Reports

┌──(kali㉿kali)-[~/University_Management/Reports/Weekly_Reports]
└─$ cd ~

┌──(kali㉿kali)-[~]
└─$
```

6- Entering a deep directory Postgraduate using an absolute path.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ cd  ~/University_Management/Students/Postgraduate
```

7- From the Postgraduate, using a relative path, list the content of the directory two level up

```
┌──(kali㉿kali)-[~/University_Management/Students/Postgraduate]
└─$ ls ../..
Attendance  Courses  Reports  Students  Tools
```

8- Entered Attendance/Postgraduate from a different path using an absolute path

```
┌──(kali㉿kali)-[~/University_Management/Students/Postgraduate]
└─$ cd  ~/University_Management/Attendance/Postgraduate
```

9- Listed the content from Postgraduate to one level up.

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Postgraduate]
└─$ ls ../
Postgraduate  Undergraduate
```

10- Confirming the current working directory location.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ pwd
/home/kali/University_Management
```

5.0 Task 3: File Manipulation & Organization

1- Renamed file Week1 without entering its directory using absolute path, and verify.

```
┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls /home/kali/kali_lab_project/Attendance/Undergraduate/
week1.txt

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ mv /home/kali/kali_lab_project/Attendance/Undergraduate/week1.txt /home/kali/kali_lab_project/Attendance/Undergr
aduate/change.txt  #Rename a file (without entering directory)


┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls /home/kali/kali_lab_project/Attendance/Undergraduate/
change.txt
```

2- Duplicated (copied) file Assignment content to a different directory Report using relative path and verified it.

```
┌──(kali㉿kali)-[~/kali_lab_project/Courses/CS101]
└─$ ls
Assignments  Grades  Syllabus.txt

┌──(kali㉿kali)-[~/kali_lab_project/Courses/CS101]
└─$ cp -r Assignments ../Report # Duplicate (copy) a file to another directory

┌──(kali㉿kali)-[~/kali_lab_project/Courses/CS101]
└─$ ls ../
CS101  MATH202  Report
```

```
┌──(kali㉿kali)-[~/University_Management/Courses/Report]
└─$ ls
Assignment1.txt  Assignment2.txt
```

3- Moved file Week1 between directories using relative path and then verified it.

```
┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ ls
week1.txt

┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ mv week1.txt ../Undergraduate #Move files between directories

┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ ls

┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ ls ../Undergraduate
change.txt  week1.txt
```

4- Moved file week1 into a new directory created earlier from the Undergraduate directory while working from a different directory using relative path, then verified it.

```
┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ mv ../Undergraduate/week1.txt backup

┌──(kali㉿kali)-[~/kali_lab_project/Attendance/Postgraduate]
└─$ ls backup
week1.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Undergraduate]
└─$ ls
change.txt
```

5- Removed an unnecessary file unes_file and an empty directory emp_dir using relative path and verified the deletion.

```
┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ ls

Automation_Scripts  emty_dir  unes_file.txt

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$  rm unes_file.txt # Remove an unnecessary file


┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ ls
Automation_Scripts  emty_dir

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$
```

```
┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ rm -r emty_dir #Remove an empty directory

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ ls
Automation_Scripts
```

6.0 Task 4: Permissions & Ownership Management

1- Checked the permissions for all directories and root directory

```
drwxrwxrwx 8 kali kali 4096 Dec 25 05:34 University_Management
```

```
┌──(kali㉿kali)-[~/University_Management]
└─$ ls -l
total 20
drwxrwxr-x 4 kali kali 4096 Dec 27 02:37 Attendance
drwxrwxr-x 5 kali kali 4096 Dec 27 02:37 Courses
drwxrwxr-x 4 kali kali 4096 Dec 25 04:29 Reports
drwxrwxr-x 4 kali kali 4096 Dec 25 04:19 Students
drwxrwxr-x 5 kali kali 4096 Jan  5 05:28 Tools
```

2- Set permissions for the University_Management directory using absolute path.

7: owner has full access (rwx)
5: group can view and enter directory, but cannot modify(r-x)
5: others same as group (r-x)

```
┌──(kali㉿kali)-[~]
└─$ chmod 755 ~/University_Management
```

```
drwxr-xr-x 8 kali kali 4096 Dec 25 05:34 University_Management
```

3- Changed directory permissions using a relative path then verified it.

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ chmod 700 Reports  #Allowing  only owner to access report

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls -l
total 20
drwxrwxr-x 4 kali kali 4096 Dec 25 09:50 Attendance
drwxrwxr-x 5 kali kali 4096 Dec 25 11:38 Courses
drwx------ 4 kali kali 4096 Dec 25 09:50 Reports
drwxrwxr-x 4 kali kali 4096 Dec 25 09:50 Students
drwxrwxr-x 3 kali kali 4096 Dec 26 00:18 Tools
```

4- Applied permission changes recursively to Students directory and its contents using relative path, and verified the change.

7: owner has full access (rwx)
5: group can view and enter directory, but cannot modify(r-x)
0: no permissions. Cannot read, write, or enter. (---)

```
┌──(kali㉿kali)-[~/kali_lab_project]
└─$ chmod -R 750 Students # owner all permision and user read and ex permision. none per other

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls -l Students
total 8
drwxr-x--- 2 kali kali 4096 Dec 25 09:50 Postgraduate
drwxr-x--- 2 kali kali 4096 Dec 25 09:50 Undergraduate

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls -l
total 20
drwxrwxr-x 4 kali kali 4096 Dec 25 09:50 Attendance
drwxrwxr-x 5 kali kali 4096 Dec 25 11:38 Courses
drwx------ 4 kali kali 4096 Dec 25 09:50 Reports
drwxr-x--- 4 kali kali 4096 Dec 25 09:50 Students
drwxrwxr-x 3 kali kali 4096 Dec 26 00:18 Tools
```

5- Changed Attendance/Undergraduate files permission using absolute path and then verified it using ls.

6: owner read and write (rw-)

4: group can read only (r–)
0: others have no access(---)

```
┌──(kali㉿kali)-[~]
└─$ chmod 640 ~/University_Management/Attendance/Undergraduate/*.txt
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance/Undergraduate]
└─$ ls -l
total 4
-rw-r——— 1 kali kali 65 Dec 27 07:04 change.txt
```

6- Changed Undergraduate directory permission using absolute path and then verified it using ls.

7: owner has full access (rwx)
5: group can view and enter directory, but cannot modify(r-x)
0: no permissions. Cannot read, write, or enter. (---)

```
┌──(kali㉿kali)-[~]
└─$ chmod 750 ~/University_Management/Attendance/Undergraduate
```

```
┌──(kali㉿kali)-[~/University_Management/Attendance]
└─$ ls -l
total 8
drwxrwxr-x 2 kali kali 4096 Jan  5 05:00 Postgraduate
drwxr-x—— 2 kali kali 4096 Jan  5 05:18 Undergraduate
```

7.0 Task 5: Exploration Tasks (Self-Directed Learning)

7.1 Example 3: Task Scheduling

1- Created a backup script file using absolute path and verified it.

```
┌──(kali㉿kali)-[~/kali_lab_project]
└─$ touch   /home/kali/kali_lab_project/backup_reports.sh # creating the backup script

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls
Attendance  backup_reports.sh  Courses  Reports  Students  Tools
```

2- Edited the script file to include the backup logic, using an absolute path.

```
┌──(kali㉿kali)-[~/kali_lab_project]
└─$ nano /home/kali/kali_lab_project/backup_reports.sh # Editing the script
```

```
kali@kali: ~/kali_lab_project
Session  Actions  Edit  View  Help
  GNU nano 8.6                    /home/kali/kali_lab_project/backup_reports.sh *
#!/bin/bash

SOURCE="/home/kali/kali_lab_project/Reports"
DEST="/home/kali/kali_lab_project/Tools/Backup"
DATE=$(date +"%Y-%m-%d_%H-%M-%S")

mkdir -p "$DEST"
tar -czf "$DEST/reports_backup_$DATE.tar.gz" "$SOURCE"
```

3- Made the script executable using chmod using an absolute path and verified it.

rwx (owner): can read, write, execute
rwx (group): can read, write, execute
r-x (others): read and execute, but no write

```
┌──(kali㉿kali)-[~/kali_lab_project]
└─$ chmod +x /home/kali/kali_lab_project/backup_reports.sh # Giving script exe permision

┌──(kali㉿kali)-[~/kali_lab_project]
└─$ ls -l /home/kali/kali_lab_project/backup_reports.sh

-rwxrwxr-x 1 kali kali 213 Dec 26 09:31 /home/kali/kali_lab_project/backup_reports.sh
```

4- Manually tested the script to ensure it runs correctly, using absolute paths.

```
kali@kali: ~/kali_lab_project/Tools
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ /home/kali/kali_lab_project/backup_reports.sh # testing the script

tar: Removing leading `/' from member names

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ ls /home/kali/kali_lab_project/Tools/Backup

reports_backup_2025-12-26_09-52-47.tar.gz
```

5- Opened the crontab editor and added a cron job to run the script every 1 minute for testing purposes.

Crontab is a Linux scheduling tool used to automatically run commands or scripts at specified times or intervals. It works by defining scheduled tasks (cron jobs) that the system executes in the background without user intervention. We used crontab in this project to automate recurring tasks, such as running a backup script, so they execute consistently and on time without needing to be run manually.

```
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
*/1 *  * * * echo "sampro22" >> /home/kali/sampro/sampro.txt
* * * * * /home/kali/kali_lab_project/backup_reports.sh
```

6- Verified that the cron job was active and successfully executed after one minute.

```
┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ crontab -e
crontab: installing new crontab

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
*/1 *  * * * echo "sampro22" >> /home/kali/sampro/sampro.txt
* * * * * /home/kali/kali_lab_project/backup_reports.sh
```

7- Verifying after some minutes using an absolute path.

```
┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ date
Fri Dec 26 10:02:45 AM PST 2025

┌──(kali㉿kali)-[~/kali_lab_project/Tools]
└─$ ls /home/kali/kali_lab_project/Tools/Backup

reports_backup_2025-12-26_09-52-47.tar.gz  reports_backup_2025-12-26_10-02-01.tar.gz
reports_backup_2025-12-26_10-00-01.tar.gz  reports_backup_2025-12-26_10-03-01.tar.gz
reports_backup_2025-12-26_10-01-01.tar.gz
```

7.2 Example 1: File Exploration and Content Analysis

1- Viewed the entire directory hierarchy without changing directories, utilizing the recursive list command. Used only relative paths for this whole task.

```
┌──(kali㉿kali)-[~]
└─$ tree University_Management

University_Management
├── Attendance
│   ├── Postgraduate
│   │   └── backup
│   │       └── week1.txt
│   └── Undergraduate
│       └── change.txt
├── Courses
│   ├── CS101
│   │   ├── Assignments
│   │   │   ├── Assignment1.txt
│   │   │   └── Assignment2.txt
│   │   ├── Grades
│   │   │   ├── student_001.csv
│   │   │   └── student_002.csv
│   │   └── Syllabus.txt
│   ├── MATH202
│   │   ├── Assignments
│   │   │   └── Assignments.txt
│   │   ├── Grades
│   │   │   └── student_101.csv
│   │   └── Syllabus.txt
│   └── Report
│       ├── Assignment1.txt
│       └── Assignment2.txt
├── Reports
│   ├── Semester_Reports
│   │   └── report_semester1.txt
│   └── Weekly_Reports
│       └── report_week1.txt
├── Students
│   ├── Postgraduate
│   │   ├── student_101.txt
│   │   └── student_102.txt
│   └── Undergraduate
│       ├── student_001.txt
│       ├── student_002.txt
│       └── student_003.txt
└── Tools
    ├── Automation_Scripts
    │   └── generate_reports.sh
    └── Backup
        ├── reports_backup_2025-12-26_09-52-47.tar.gz
        ├── reports_backup_2025-12-26_10-00-01.tar.gz
        ├── reports_backup_2025-12-26_10-01-01.tar.gz
        ├── reports_backup_2025-12-26_10-02-01.tar.gz
        ├── reports_backup_2025-12-26_10-03-01.tar.gz
        ├── reports_backup_2025-12-26_10-04-01.tar.gz
        ├── reports_backup_2025-12-26_10-05-01.tar.gz
        ├── reports_backup_2025-12-26_10-06-01.tar.gz
        ├── reports_backup_2025-12-26_10-07-01.tar.gz
        ├── reports_backup_2025-12-26_10-08-01.tar.gz
        ├── reports_backup_2025-12-26_10-09-01.tar.gz
        └── reports_backup_2025-12-26_10-10-01.tar.gz

22 directories, 32 files

┌──(kali㉿kali)-[~]
```

2- Searched for all text files within the project directory

```
┌──(kali㉿kali)-[~]
└─$ find University_Management -name "*.txt"
University_Management/Attendance/Postgraduate/backup/week1.txt
University_Management/Attendance/Undergraduate/change.txt
University_Management/Courses/MATH202/Syllabus.txt
University_Management/Courses/MATH202/Assignments/Assignments.txt
University_Management/Courses/Report/Assignment1.txt
University_Management/Courses/Report/Assignment2.txt
University_Management/Courses/CS101/Syllabus.txt
University_Management/Courses/CS101/Assignments/Assignment1.txt
University_Management/Courses/CS101/Assignments/Assignment2.txt
University_Management/Reports/Weekly_Reports/report_week1.txt
University_Management/Reports/Semester_Reports/report_semester1.txt
University_Management/Students/Postgraduate/student_101.txt
University_Management/Students/Postgraduate/student_102.txt
University_Management/Students/Undergraduate/student_001.txt
University_Management/Students/Undergraduate/student_003.txt
University_Management/Students/Undergraduate/student_002.txt

┌──(kali㉿kali)-[~]
└─$
```

.

3- Identified all CSV files within University_Management.

```
┌──(kali㉿kali)-[~]
└─$ find University_Management -type f -name "*.csv"
University_Management/Courses/MATH202/Grades/student_101.csv
University_Management/Courses/CS101/Grades/student_002.csv
University_Management/Courses/CS101/Grades/student_001.csv
```

4- Verified the specific file type of the 'student_101.csv' file.

```
┌──(kali㉿kali)-[~]
└─$ file University_Management/Courses/MATH202/Grades/student_101.csv
University_Management/Courses/MATH202/Grades/student_101.csv: ASCII text
```

5- Opened and edited file Assignment.txt using the nano text editor to add content. And verified using cat.

```
┌──(kali㉿kali)-[~/University_Management/Courses/MATH202]
└─$ nano Assignments/Assignments.txt
```

6- Verified used cat.

```
┌──(kali㉿kali)-[~/University_Management/Courses/MATH202/Assignments]
└─$ cat Assignments.txt
This file is for the math subject assignments
Im adding a new line of text. Student Eylaf
```

7- View content of change.txt file using cat.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ cat  Attendance/Undergraduate/change.txt
im adding text to change file
```

8- Added text using the vim editor for advanced editing to file change.txt

```
┌──(kali㉿kali)-[~/University_Management]
└─$ vim  Attendance/Undergraduate/change.txt
```

9- Verified using cat.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ cat  Attendance/Undergraduate/change.txt
im adding text to change file
line 2
line 3
line 4
line 5
line 6
```

10- Used the less command to view the content of a file one page at a time.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ less Attendance/Undergraduate/change.txt
```

```
im adding text to change file
line 2
line 3
line 4
line 5
line 6
Attendance/Undergraduate/change.txt (END)
```

9- Displayed the first 3 lines of a file to preview its content.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ head -n 3 Attendance/Undergraduate/change.txt
im adding text to change file
line 2
line 3
```

10- Searched all files for text “Eylaf” and suppressed the binary files.

```
┌──(kali㉿kali)-[~]
└─$ grep -R -I "Eylaf" University_Management

University_Management/Courses/MATH202/Assignments/Assignments.txt:Im adding a new line of text. Student Eylaf
```

11- Counted the number of lines, words, and characters in a file.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ wc  Attendance/Undergraduate/change.txt
 6 16 65 Attendance/Undergraduate/change.txt
```

12- Retrieved detailed information including timestamps and permission information for subdirectory Students.

```
┌──(kali㉿kali)-[~/University_Management]
└─$ stat Students
  File: Students
  Size: 4096            Blocks: 8          IO Block: 4096   directory
Device: 8,1     Inode: 3539411     Links: 4
Access: (0750/drwxr-x───)  Uid: ( 1000/    kali)   Gid: ( 1000/    kali)
Access: 2026-01-05 08:00:57.926602852 -0500
Modify: 2025-12-25 04:19:03.022503745 -0500
Change: 2026-01-05 08:00:57.922602904 -0500
 Birth: 2025-12-25 04:11:18.784672663 -0500
```

8.0 Challenges and Learning Outcomes

During this project, challenges included planning a logical directory hierarchy, handling file and folder operations from different levels, managing permissions correctly, and resolving conflicts when collaborating with Git. These challenges helped improve proficiency in CLI commands, file organization, version control with Git, and automation using scripts. By completing the tasks, a strong understanding of operating system fundamentals and practical command-line workflows was gained.

9.0 Member Involvement

Eylaf Mahdi Adam: Completed Task 1 (Directory & Project Structure Creation), Task 2 (Directory Navigation Using Paths), and Example 1 of Task 5 (File Exploration and Content Analysis). Additionally, contributed to preparing the presentation slides and assisted in writing and reviewing the final report.

Sankumba Balajo: Completed Task 3 (File Manipulation & Organization), Task 4 (Permissions & Ownership Management), and Example 3 of Task 5 (Automation or Task Scheduling). Also contributed to preparing the report.

10.0 Conclusion:

This project provided hands-on experience in managing files, directories, and permissions using the Kali Linux command line. Through tasks such as directory creation, navigation, file manipulation, permission management, and automation, we strengthened our understanding of operating system fundamentals and CLI functionality. The exploration tasks further enhanced our ability to analyze and organize files efficiently, while automating repetitive actions demonstrated practical applications of scripting. Overall, the project improved our technical proficiency, problem-solving skills, and ability to work collaboratively in a structured computing environment.